

Comparison of Real-Time Object Detection Models on Webcam-Based Video Input

1st Ayden Frometa
Department of Computer Science
Florida Polytechnic University
Lakeland, United States
afrometa4253@floridapoly.edu

2nd Payne Persons
Department of Computer Science
Florida Polytechnic University
Lakeland, United States
ppersons4120@floridapoly.edu

Abstract—This paper presents an experimental comparison of three object detection models: YOLOv8, MobileNet-SSD, and EfficientDet. The models are evaluated on webcam-based video input to compare their real-time performance and detection-output behavior under realistic conditions. The evaluation uses frames per second, latency, average confidence, and number of detections as the primary comparison metrics. Results show that MobileNet-SSD achieves the highest real-time speed, YOLOv8 provides the strongest balance between speed and detection activity, and EfficientDet is not practical for real-time webcam use on the tested system because of its high latency.

Index Terms—object detection, YOLOv8, MobileNet-SSD, EfficientDet, real-time detection, webcam benchmarking

I. INTRODUCTION

A. Background and Significance

Object detection is an important area of computer vision that has advanced rapidly with the development of deep learning models. Different object detection models use different architectures and inference strategies, which can lead to differences in speed, accuracy, and real-time usability. Real-time detection is especially important for applications that require immediate feedback from a video stream. The real-world performance of object detection models on webcam-based video input may differ from performance on static datasets because of environmental variability.

B. Motivation

Webcam-based environments can introduce several challenges:

- Multiple objects
- Cluttered backgrounds
- Motion blur
- Lighting variation
- Webcam placement

When these elements are factored in, studying how different models perform is essential for selecting appropriate detectors for real-time systems.

C. Problem Definition

Question: Which object detection model performs best on webcam-based video input?

Performance is assessed on the basis of detection quality and real-time efficiency.

D. Overview of Related Work

Deep learning has significantly advanced real-time object detection. The YOLO framework introduced a unified single-stage architecture that enabled fast detection while maintaining competitive accuracy [1]. Modern implementations such as YOLOv8 continue to emphasize the balance between speed and detection performance for real-time applications [2].

Lightweight detectors such as MobileNet-SSD are designed for resource-constrained environments and typically prioritize inference speed and efficiency [3]. In contrast, EfficientDet improves the accuracy–efficiency trade-off through compound scaling and the bidirectional feature pyramid network (BiFPN) [4]. Comparative studies show that these models exhibit different performance characteristics depending on computational constraints and scene complexity [5].

However, most prior evaluations focus on benchmark datasets rather than webcam-based environments with real-world variability. This work provides an experimental comparison of representative real-time detectors under webcam-based conditions using recorded webcam footage for controlled benchmarking.

II. EXPLANATION OF EXPERIMENT

This project performs an experimental comparison of three widely used object detection models: YOLOv8, MobileNet-SSD, and EfficientDet. Each model represents a different design philosophy in real-time object detection. YOLOv8 represents a balanced real-time detector designed for both speed and accuracy. MobileNet-SSD serves as a lightweight, resource-efficient baseline suitable for real-time detection tasks. EfficientDet represents an accuracy-focused model that uses compound scaling and a bidirectional feature pyramid network, although it runs more slowly than the other models on the tested system, especially on CPU.

The experiment uses recorded webcam footage to ensure that each model processes the same input under identical conditions. A webcam video is first recorded and saved as a video file. Each model is then run sequentially on the same footage, allowing performance differences to be attributed to the models rather than changes in the input stream. This approach provides a controlled comparison while still reflecting realistic webcam-based object detection conditions.

The benchmarking pipeline consists of video capture, frame preprocessing, model inference, detection output collection, and metric logging. For each processed frame, the system records performance metrics such as frames per second and latency, along with detection-output metrics such as average confidence and number of detections. Precision, recall, F1 score, and mean Average Precision (mAP) are included in the output structure, but they require ground-truth bounding boxes. Since the webcam footage is not manually annotated, these values remain zero, and average confidence and number of detections are used as practical detection-quality indicators.

III. EVALUATION

The setup for this project was done according to the instructions in the README file. They are as follows.

The required Python dependencies were installed using the project requirements file, and the MobileNet-SSD architecture and weight files were downloaded separately. The YOLOv8 weights download automatically during the first run, while EfficientDet is loaded from TensorFlow Hub. A webcam recording was then created using the recording script and saved as a video file before running the benchmark script on the recorded footage. This setup ensured that each model was evaluated using the same input video and that the results were saved consistently in CSV format.

The project evaluates the performance of YOLOv8, MobileNet-SSD, and EfficientDet using both real-time performance metrics and detection-output metrics. Each model is tested on webcam-based input so that its practical usability for real-time object detection can be compared. The evaluation uses the benchmark output saved in CSV format, where each row records the results for one processed frame.

The primary real-time performance metrics are frames per second (FPS) and per-frame latency. FPS measures how many video frames the model can process each second, while latency measures the amount of time required to process a single frame. A higher FPS and lower latency indicate better real-time performance. According to the project README, 24 FPS is used as a general threshold for smooth real-time performance, which corresponds to a latency of approximately 42 milliseconds per frame.

Detection quality is evaluated using the average confidence score and the number of detections per frame. Average confidence measures the model's confidence in its detections, while the number of detections shows how many objects the model identifies in each frame. The benchmark also includes precision, recall, F1 score, and mean Average Precision (mAP); however, these metrics require ground-truth bounding boxes. Since the webcam footage is not manually annotated, these values are not used as the main basis for comparing detection quality.

Based on the recorded-video benchmark, MobileNet-SSD achieved the highest average speed at approximately 24.53 FPS with an average latency of 41.61 ms, making it the closest to smooth real-time performance. YOLOv8 achieved approximately 19.86 FPS with an average latency of 51.52

ms, which is slower than MobileNet-SSD but still much faster than EfficientDet. EfficientDet achieved only approximately 0.28 FPS with an average latency of 3674.17 ms, showing that it is not practical for real-time webcam use on the tested system.

For detection-output behavior on the recorded video, YOLOv8 produced an average confidence score of approximately 0.74 and detected about 2.86 objects per frame. MobileNet-SSD produced a slightly higher average confidence score of approximately 0.75, but detected fewer objects, averaging about 1.95 detections per frame. EfficientDet produced an average confidence score of approximately 0.64 and the highest number of detections, averaging about 3.94 detections per frame, but its extremely low FPS makes it unsuitable for real-time use.

The live-feed benchmark showed similar performance trends. YOLOv8 ran at approximately 9.87 FPS with 105.45 ms latency, while MobileNet-SSD ran at approximately 10.62 FPS with 96.01 ms latency. EfficientDet again performed much slower, reaching only approximately 0.29 FPS with an average latency of 4863.23 ms. These results indicate that EfficientDet is the least practical model for real-time webcam detection on the tested system.

Overall, MobileNet-SSD provides the best real-time speed, especially in the recorded-video benchmark, where it nearly reaches the 24 FPS target. YOLOv8 provides a strong balance between speed and detection activity, detecting more objects per frame than MobileNet-SSD while maintaining usable performance. EfficientDet detects the most objects per frame in the recorded-video test, but its very high latency prevents it from being considered a practical real-time option. Therefore, MobileNet-SSD is the strongest choice for speed-focused real-time use, while YOLOv8 is the best balanced option when both speed and detection output are considered.

IV. CSV BENCHMARK RESULTS

The benchmark results saved in the CSV files provide a frame-by-frame record of each model's performance on both recorded webcam video and live webcam input. Each CSV row represents one processed frame and includes the model name, frame number, FPS, latency, precision, recall, F1 score, mAP, average confidence, and number of detections. Because the webcam footage was not manually annotated with ground-truth bounding boxes, precision, recall, F1 score, and mAP are not used as the main basis for evaluating detection quality. Instead, the analysis focuses on FPS, latency, average confidence, and number of detections.

Table I summarizes the average results from the recorded-video CSV file. MobileNet-SSD achieved the highest average speed at 24.53 FPS and the lowest average latency at 41.61 ms. This makes it the strongest model for speed-focused real-time performance in the recorded-video test. YOLOv8 reached 19.86 FPS with an average latency of 51.52 ms, making it slower than MobileNet-SSD but still usable for real-time or near-real-time applications. EfficientDet performed much more slowly, averaging only 0.28 FPS with a latency of 3674.17 ms,

which makes it unsuitable for real-time webcam detection on the tested system.

TABLE I
AVERAGE RECORDED-VIDEO CSV RESULTS

Model	FPS	Latency (ms)	Avg. Conf.	Avg. Detections
YOLOv8	19.86	51.52	0.74	2.86
MobileNet-SSD	24.53	41.61	0.75	1.95
EfficientDet	0.28	3674.17	0.64	3.94

The recorded-video results show that MobileNet-SSD is the fastest model, while YOLOv8 provides stronger detection activity than MobileNet-SSD. YOLOv8 detected an average of 2.86 objects per frame, compared with 1.95 objects per frame for MobileNet-SSD. EfficientDet detected the most objects per frame, with an average of 3.94 detections, but this result must be interpreted together with its extremely poor speed. A model that detects more objects but takes several seconds per frame is not practical for real-time webcam use.

Table II summarizes the average results from the live-feed CSV file. The live-feed results show a similar overall trend, although both YOLOv8 and MobileNet-SSD were slower than they were in the recorded-video benchmark. YOLOv8 averaged 9.87 FPS with 105.45 ms latency, while MobileNet-SSD averaged 10.62 FPS with 96.01 ms latency. EfficientDet again performed far below real-time speed, averaging only 0.29 FPS with 4863.23 ms latency.

TABLE II
AVERAGE LIVE-FEED CSV RESULTS

Model	FPS	Latency (ms)	Avg. Conf.	Avg. Detections
YOLOv8	9.87	105.45	0.54	2.03
MobileNet-SSD	10.62	96.01	0.04	0.12
EfficientDet	0.29	4863.23	0.49	3.32

The live-feed CSV results further support the conclusion that EfficientDet is not practical for this webcam-based real-time detection setup. Although EfficientDet produced more detections per frame than the other models, its latency was several thousand milliseconds per frame. MobileNet-SSD remained the fastest model in the live-feed test, but its average confidence and number of detections were much lower than YOLOv8. YOLOv8 was slightly slower than MobileNet-SSD, but it produced more detections per frame and a higher average confidence score.

Overall, the CSV results show that MobileNet-SSD is the best option when speed is the highest priority. YOLOv8 is the strongest balanced option because it maintains reasonable speed while producing more consistent detection output. EfficientDet produces a relatively high number of detections, but its very low FPS and high latency make it unsuitable for real-time webcam-based object detection on the tested system.

V. CONCLUSION

This work presented an experimental comparison of YOLOv8, MobileNet-SSD, and EfficientDet on webcam-based video input. By evaluating each model with real-time performance metrics and detection-output metrics, the project identified important trade-offs between speed, latency, confidence, and number of detections. The results showed that MobileNet-SSD achieved the strongest real-time speed, YOLOv8 provided the best overall balance between performance and detection activity, and EfficientDet was not practical for real-time use on the tested system due to its high latency. Because the webcam footage was not manually annotated with ground-truth bounding boxes, the detection-quality comparison is based on output behavior rather than precision, recall, F1 score, or mAP. Overall, the study demonstrates that model selection for webcam-based object detection depends on whether the priority is maximum speed or a balance between speed and detection output.

REFERENCES

- [1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," 2016. [Online]. Available: <https://arxiv.org/abs/1506.02640>
- [2] Ultralytics, "Yolov8 models documentation," <https://docs.ultralytics.com/models/yolov8/>, 2023, accessed: 2026.
- [3] S. Islam, G. Ferraioli, V. Pascasio, S. Vitale, and M. Amin, "Performance analysis of yolov3, yolov4 and mobilenet ssd for real time object detection," vol. 5, pp. 37–49, 06 2024.
- [4] M. Tan, R. Pang, and Q. V. Le, "Efficientdet: Scalable and efficient object detection," 2020. [Online]. Available: <https://arxiv.org/abs/1911.09070>
- [5] M. I. Khalif and R. D. Mardian, "Comparative analysis of yolov3, mobilenet-ssd, and efficientdet for real-time person detection in low-resolution images," *ITEGAM- Journal of Engineering and Technology for Industrial Applications (ITEGAM-JETIA)*, vol. 11, 01 2025.